

Antonio Coppola, Ph.D.

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Software and Optimization Engineer with 4+ years of experience developing algorithmic solutions and software for large-scale operational problems. Strong background in Python, C++, TypeScript, algorithms, and mathematical optimization, combined with experience in machine learning, data pipelines, CI/CD, and real-world system implementation and validation. Experienced in developing, benchmarking, and deploying solutions for complex, data-intensive applications.

EXPERIENCE

Optimization Engineer — Transit Systems

2024–2025

MCube Cluster (Munich Cluster for the Future of Mobility) | STEAM Project

- Developed an optimization system end-to-end, from problem definition and algorithm design through implementation and field testing.
- Implemented algorithmic solutions for real operational requirements in collaboration with BMW, SWM, and MAN.
- Developed and validated software and data pipelines using GPS telemetry from real-world operations.
- Developed demand-responsive scheduling logic that reduced walking distances by 20% and increased vehicle occupancy by 15%.

Optimization & AI Engineer — Intelligent Mobility

2022–2026

Technical University of Munich | supervised by Prof. Maximilian Schiffer

- Developed scalable algorithmic solutions for complex mobility problems across the software lifecycle: data, modeling, implementation, experimentation, and deployment.
- Implemented optimization algorithms combining matheuristics, metaheuristics, and machine learning.
- Reduced congestion-related delays by up to 35% versus baseline under realistic demand.
- Developed and maintained software and data pipelines in Python, Julia, and C++ using Git and CI/CD workflows.
- Benchmarked solution quality, runtime, and scalability of algorithmic methods.
- Developed ML models and GenAI/LLM workflows for algorithmic and data-driven applications.
- Communicated technical results and solution approaches to academic and industry stakeholders.

Optimization Engineer — Vehicle Routing

2025

Polytechnique Montréal | supervised by Prof. Michel Gendreau

- Implemented and benchmarked Large Neighborhood Search algorithms for real-world vehicle routing problems.
- Evaluated algorithmic solutions in terms of solution quality and computational performance.

Systems Engineering Intern — Network Dynamics

2021

University of Naples Federico II | supervised by Prof. Franco Garofalo

- Developed analysis and modeling methods for complex networks, resulting in a peer-reviewed publication.

EDUCATION

Ph.D. in Operations Research and Machine Learning

2022–2026

Technical University of Munich, Germany

M.Sc. in Industrial Engineering (110/110 cum laude)

2018–2021

University of Naples Federico II, Italy

B.Sc. in Industrial Engineering

2014–2018

University of Naples Federico II, Italy

SKILLS

Software & Programming: Python, C++, TypeScript, Julia, Git, CI/CD, Docker, GitHub Copilot; software development, testing, deployment, performance benchmarking.

Algorithms & Optimization: algorithm development, data structures, LP/IP/MILP, metaheuristics, matheuristics, Large Neighborhood Search, network optimization; Gurobi, CPLEX.

Machine Learning & AI: GNNs, predictive modeling, PyTorch, scikit-learn, GenAI, agentic LLM workflows.

Data & Cloud: working familiarity with SQL/NoSQL and AWS; data pipelines and data-intensive applications.

Languages: Italian (native), English (C2), German (B2), Portuguese & French (B1)

PUBLICATIONS

A. Coppola et al. — *Staggered Routing in AMoD Systems*, European Journal of Operational Research, 2025

A. Coppola et al. — *Integrated Balanced and Staggered Routing in AMoD Systems*, under review, 2026

A. Coppola et al. — *Online Balanced and Staggered Routing in AMoD Systems*, under review, 2026

C. Ancona, ..., A. Coppola et al. — *Partial Controllability of Network Dynamical Systems*, IEEE Control Systems Letters, 2021

FELLOWSHIPS & FUNDING

Bavaria–Québec Mobility Program — BayFOR

2025

Academic Training Program — TUM School of Management

2024

Research Fellowship — University of Naples Federico II

2021

Erasmus+ Program — European Commission

2019